

Governance and Conflict Relapse

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Abstract

Many conflict studies regard formal democratic institutions as states' most important vehicle to reduce deprivation-motivated armed conflict against their governments. We argue that the wider concept of *good governance*—the extent to which policy making and implementation benefit the population at large—is better suited to analyze deprivation-based conflict. The article shows that the risk of conflict in countries characterized by good governance drops rapidly after a conflict has ended or after independence. In countries with poor governance, this process takes much longer. Hence, improving governance is important to reduce the incidence of conflict. We also decompose the effect of good governance into what can be explained by formal democratic institutions and less formal aspects of governance, and into what comes from economic development and what is due to how well countries are governed. We find that informal aspects of good governance to be at least as important as formal institutions in preventing conflict and that good governance has a clear effect over and beyond economic development.

Keywords

civil war, conflict recurrence, governance, democracy, institutions

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Are well-governed countries better able to avoid armed conflict? Studies such as Hegre et al. (2001), Fearon and Laitin (2003), and Cederman, Hug, and Krebs (2010) answer this question focusing on the extent to which the governing institutions are *formally democratic* since democratically elected leaders should be best placed to address grievances. The rules regulating how political leaders are recruited and how citizens participate in this selection, however, are not the only relevant determinants of “good governance.” Potentially, the quality of the bureaucratic apparatus,¹ the extent of political corruption, and the appropriateness of the economic policies chosen by political leaders also affect governments’ ability to prevent domestic violent conflicts. In this article, we analyze all country-years from 1960 through 2008 and study how such an expanded conception of governance might increase our understanding of armed conflict recurrence.²

The article first looks at the literature treating theoretical links between governance and conflict, focusing on relative deprivation (the Theoretical Background section). The Measuring Governance section discusses briefly the measures we use to capture the concept of *good governance*. In addition to identifying seven disaggregated indicators, we combine these into a joint governance index and analyze how traditional democracy indicators relate to more informal concepts of governance such as corruption and bureaucratic quality. Next, we analyze the effect of governance on conflict (the Governance and Conflict section).

Our statistical models show that the risk of conflict in countries with good governance drops rapidly after the conflict has ended and after independence. In countries characterized by poor governance, this process takes much longer. Hence, improving governance plays an important role in reducing the incidence of conflict.

We find most of our disaggregated governance indicators to be related to a decreased risk of conflict recurrence. This is partly due to positive correlation between the various indicators, but also it suggests that reform in the different sectors can be *partial substitutes* to each other. In some cases, it is a challenge to find an entry point to break this vicious circle of poor governance and conflict. If reform in any sector helps, changes should be sought where they are most feasible.

We also analyze to what extent good governance can be separated from formal democratic institutions on one hand, and economic development on the other. We show that a model that takes both dimensions of governance into account fit the data better than a model that only looks at one of these two dimensions. Both aspects of governance are important for preventing conflict recurrence, whereas informal governance seems more important for preventing conflict onset. Likewise, we find a model that separates the two dimensions of governance from average income to improve the fit to the data. In particular, countries that are relatively democratic for their income levels but score poorly on informal governance (e.g., Gabon or Lebanon) are at risk of conflict.

Theoretical Background

Arguments relating to the relationship between aspects of governance and the risk of internal armed conflict typically build at least implicitly on accounts of how the psychological state of “relative deprivation” is transformed into intergroup conflict (Davies 1962; Gurr 1968). In the words of Gurr (1968, 1104), “Relative deprivation is defined as actors’ perceptions of discrepancy between their value expectations (the goods and conditions of life to which they believe they are justifiably entitled) and their value capabilities (the amounts of those goods and conditions that they think they are able to get and keep).” Davies (1962) sees relative deprivation as a dynamic phenomenon—people typically accept a gap between actual and expected “need satisfaction,” but the risk of rebellion increases if this gap starts to increase toward an intolerable level. We follow Gurr and Davies in thinking of “actors” very broadly as government elites and the nonelite majority population in a country.

We focus here solely on internal armed conflicts that pit governments against domestic opposition groups. Relative deprivation can explain internal conflicts if populations perceive governments as the source of the expectations-ability discrepancy and are willing to support armed actions to address the issue. In other words, such explanations necessarily imply there is a lack of good governance, defined as the extent to which the process of policy making and implementation of policies are to the benefit of the public at large. Gurr’s (1968) formulation of the relative-deprivation explanation indeed includes aspects of governance such as institutionalization or legitimacy of the regime. Quantitative studies of deprivation- or grievance-based explanations, however, typically tend to focus on simple “grievance factors” such as poverty, inequality, or absence of democracy. Focusing more precisely on a “lack of good governance” better reflects the relative-deprivation argument. This is the case for two reasons.

First, it requires that the government is seen as *responsible* for the “expectations-ability discrepancy.” Economic inequality is one case in point. Collier and Hoeffler (2004, 570) sees inequality as an “objective measure of grievance,” and Fearon and Laitin (2003, 79) state that “greater economic inequality creates broad grievances that favor civil conflict.” Neither of these find “grievance” such defined to have any association with a heightened risk of armed conflict. However, it is not necessarily so that governments are perceived as responsible for an income gap. Often, inequalities have much deeper historical roots than the current government. The income differences between Slovenia and other parts of former Yugoslavia may be traced back to the location of the border between the Habsburg and Ottoman empires in the eighteenth century, for instance, and differences between the various castes of India have even longer historical pedigrees.³ That inequality is unrelated to the risk of conflict may not be taken as a refutation of relative-deprivation-based explanations, since it is not an outcome of current government actions.⁴ Similarly, poverty at the country level is not a good proxy for relative deprivation (Collier and Hoeffler 2004). The fact that

most citizens in African countries are much poorer than citizens in Europe is not automatically translated into a sentiment of relative deprivation strong enough to motivate rebellion against their own governments. The natural target of such sentiments might rather be the former colonial power or other external actors.

Second, our relative-deprivation formulation relates the source of conflict to whether decision making and implementation of decisions are to the benefit of the public in practice, not necessarily to the *de jure* institutions that lay out the rules for decision making. Earlier studies (e.g., Hegre et al. 2001; Fearon and Laitin 2003; Collier and Hoeffler 2004) implicitly reduce the concept of good governance to an issue of whether the government is formally democratic or not—whether the opposition has a “voice” through which they can address grievances (Walter 2004). If selection into central political positions occur through general elections, governments are presumed to govern in ways that reduce this expectations-ability discrepancy. This, however, is insufficient. First, nonelected governments may also choose to promote policies in the interest of the general population. Moreover, elected governments may fail to govern well despite the best intentions, for instance, when an underdeveloped public sector is unable to implement policies decided within the representative institutions. Representative institutions cannot automatically be equated with “good governance.”

A more general empirical implication of the relative-deprivation conflict hypothesis is that good governance broadly defined reduces incentives for conflict due to relative deprivation. Our core research objective is to investigate whether such a link exists empirically, and to see how it relates to the formal setup of institutions. In the remainder of the Theoretical Background section, we spell out in somewhat more detail how good governance as defined previously should reduce the risk of armed conflict, concentrating on aspects of governance that are empirically observable. We then discuss the independent effect of formal democratic institutions.

How Good “Informal Institutions” Reduce Conflict

Our conception of good governance involves two distinct aspects: First, that governments intend to implement policies in the public interest and that they succeed in implementing them effectively. We outline six broad categories of “informal institutions” that affect the quality of governance, and contrast them with “formal” institutions of governance.

Several nonquantitative studies highlight the importance of certain such “prerequisites” for democratic institutions to be effective. Zakaria (1997), for instance, includes effective “rule of law” as an important element of liberal democracy. Still, there are few quantitative studies of the impact on conflict of less formalized aspects of governance, such as corruption, economic policies, or bureaucratic quality. In some cases, informal governance may be as important as formal institutions, for example, if political corruption is so rampant that elected officials are more accountable to their paymasters than to those who elected them into office.

Our conception of governance captures the entire “chain” or context of implementation that translates the aggregation of preferences into final implementations of policies. Such a wide definition of governance implies that it is a multifaceted concept, and various sources identify different dimensions of it. In this article, we look into seven aspects: (1) bureaucratic quality, (2) the rule of law, (3) corruption, (4) economic policies, (5) military involvement in politics, (6) political exclusion and repression, and (7) formal political institutions. Indicators of quality of governance along these dimensions are drawn from several different data sets.

Bureaucratic quality. Efficient bureaucracies are necessary to implement public policies and to carry out day-to-day administration. High-quality bureaucracies may also function as an informal constraint on the executive branch of the government, thereby reducing the incentives for extreme policies. With “bureaucratic quality,” we mean the capacity of the political systems to implement decisions through its administrative apparatus.⁵ There are hardly any studies of the relationship between bureaucratic quality and conflict, but several on the related concept of “state capacity.” There is, however, no consensus definition of state capacity, and there is a tendency to conflate measures of regime type and capacity.⁶

Fearon and Laitin (2003) argue that state capacity, which they proxy by per capita income, decreases the risk of onset of civil war. Other studies find that oil producers have higher risk of civil war than expected from their income levels (Hegre and Sambanis 2006, 528). Fearon (2005) argues this is because oil-producing states tend to be underbureaucratized for their gross domestic product (GDP) levels, since they have never needed to extract taxes and other resources from the citizenry at large.

Using similar data to what we employ subsequently, Fjelde and de Soysa (2009) disaggregate state capacity into the abilities to coerce, to coopt, and to cooperate. They argue that cooptation and cooperation may be more important aspects of state capacity in preventing conflict than purely coercive ability. To arrive at a better proxy for a state’s capacity, they use the “Relative Political Capacity” (RPC) index from Organski and Kugler (1980) which compares actual to predicted levels of state tax extraction. Braithwaite (2010) finds some evidence for the hypothesis that countries with good state capacity are better able to avoid spillover from neighboring countries.

Rule of law. Countries differ considerably in the extent to which judicial systems succeed in applying laws equally to all citizens. “Rule of law” is also the aspect of good governance that ensures that contracts and property rights are enforced and judiciaries are isolated from inference from decision makers. Countries characterized by such rule of law have less social conflict (Taydas, Peksen, and James 2010). This aspect of governance is related to formal democratic institutions, as they allow citizens to elect representatives to legislatures and to thereby influence lawmaking. However, many democracies such as Paraguay or Lebanon have weak rule of law, severely undermining this aspect of representation. At the same time, some

nondemocracies such as Oman or Vietnam have good judicial systems according to the International Country Risk Guide (ICRG; PRS Group 2010).

Corruption. Large-scale corruption has several detrimental effects on a political system, and several studies find corruption to be associated with political instability (Le Billon 2003; Mauro 1995; Taydas, Peksen, and James 2010). Corruption's effect on the allocation of public resources, Le Billon argues, may increase grievance and elicit demands for political change. Moreover, the rents generated by corruption may constitute a tempting prize for actors willing to use violence to capture political positions. It means a diversion of public funds, thereby decreasing public spending that reduce political conflict in the long run. Finally, corruption undermines the government's ability to implement public policies that generate economic growth and other outcomes that reduce the risk of conflict. Fjelde (2009) shows that increasing corruption from the 5th to the 95th percentile more than triples the risk of conflict onset. At the same time, she argues that corruption has no conflict-inducing effect in countries with large oil revenues. In such states, where funds for discretionary use are ample, "a strategic use of public resources for off-budget and selective accommodation of private interests might reduce the risk of violent challenges to state authority" (Fjelde 2009, 214).

Economic policies. Some policies with conflict-reducing potential can be assumed to be desirable for most decision makers, regardless of how accountable they are to the public. Policies that ensure economic growth, for instance, have been shown to reduce discontent. Strong economic growth reduces poverty even without redistribution, and even if inequality increases. This is the case in China, for instance. Positive growth is strongly related to stability of political institutions and a low risk of armed conflict (Przeworski et al. 2000; Collier and Hoeffler 2004, Gates et al. 2006).

A few studies show a direct link between economic policies and the risk of conflict. Bussmann and Schneider (2007) conclude that economic openness in the form of interstate trade and foreign direct investment (FDI) inflows decrease the risk of internal conflict, although the process of opening up the economy often is associated with increased levels of violence. Relatedly, Dreher, Gassebner, and Siemers (2012) show that economic freedom and globalization reduce countries' human rights violations. In many cases, economic policies that are inappropriate from the point of view of the public are implemented as part of "kleptocratic" strategies designed to enrich narrow elites. Accordingly, sound economic policies are likely to reduce the risk of conflict even if growth is weak due to external market conditions.

Military in politics. Heavy military involvement in government may lead to poor governance, particularly as pertains to the risk of conflict relapse. Military regimes, on average, tend to have poor rates of economic performance (Wintrobe 1998; Knutsen 2011) and have the lowest life expectancy of any type of regime (Geddes 1999).

Militaries, moreover, are not accountable to the public and are more likely to distort public spending in favor of military spending than civilian actors, diverting funds from policies that may reduce the long-term risk of conflict. In many cases, militaries seek military solutions to conflict or they invite militarized challenges to their power positions.

Political exclusion and repression. Systematic exclusion from decision-making bodies is likely to turn people against them (Cederman, Wimmer, and Min 2010). Such exclusion is particularly likely to lead to organized mass opposition when exclusion is in terms of ethnic and religious groups, since such groups typically have networks that facilitate organization and the exclusion is in terms of preexisting identity markers. Repressive policies reinforce such exclusion and accentuate group conflicts.

Political exclusion is obviously intrinsically linked to the indicators of formal institutions discussed subsequently—democratic institutions should in principle give every individual equal votes. But in practice democratic institutions may also exclude citizens, either because they limit citizen rights such as is the case for the territories occupied by Israel, or because permanent minorities in national assemblies in practice are excluded from government positions. On other hand, the rules of exclusion in some nondemocratic systems are independent of ethnic or religious cleavages in society.

We also include countries' respect for human rights as one aspect of good governance. It is difficult to estimate the direct consequence of repression on the risk of conflict, however, as dissent, repression, and poor economic performance are heavily intertwined (Moore 1998; Carey 2007; Davenport 2007). Repression increases the risk of political violence, but repression also increases when armed conflict breaks out. Some regimes, though, are clearly able to sustain repressive policies for decades. This ability is likely connected with state capacity, as discussed previously. Just as high-capacity or resource-rich regimes have ample funds to co-opt the opposition, they are able to employ repression without increasing the chances of regime collapse (Fjelde and de Soysa 2009).

Formal Institutions

With "formal institutions," we mean *de jure* institutions that ensure that the executive branch of government is elected by a majority or plurality of the population and that guarantee that the executive is constrained by an elected legislature. The academic consensus of the formal institution–conflict relationship based on global comparisons is that democracy in itself does not reduce the risk of civil war onset—if we take countries' income into account, democracies are no less likely to have internal conflicts than nondemocracies (Muller and Weede 1990; Hegre et al. 2001; Fearon and Laitin 2003; Collier and Hoeffler 2004).⁷ Moreover, semi-democracies—regimes that are partly democratic, partly autocratic such as Lebanon in the 1970s

or Tanzania today—may even have the highest risk of civil war onset. Democracies are not in a better position to end conflicts (Fearon 2004), nor is there any conclusive evidence that democratic institutions do better in postconflict settings than they do in general.⁸

A few studies note qualifications to the conclusion that formal political institutions have no conflict-reducing effects. Hegre (2003) and Collier and Rohner (2008) note that democratic institutions are particularly ineffective in low-income countries. In the poorest countries in the world, democratic institutions may even increase the risk of insurrections. In middle-income countries, these studies indicate that democratic political systems are better able to prevent conflicts from erupting and recurring. Formally democratic political institutions are clearly not sufficient in situations where governments have only partial control over own territory, and therefore are easily challenged by narrow groups. But the inability of low-income democracies to prevent armed conflicts may also be related to weak or inappropriate *informal* institutions of governance.

Democratic governments ideally succeed in reducing deprivation-based social discontent since the executive and the legislature is recruited through free and fair elections and thereby represent the “will of the people.” In practice, however, elected decision makers must abide by their own laws for this to happen. The military, or actors with the means to grease politicians, should be unable to influence decisions in their own favor. And, even if decisions are made with the sincere intention to reduce poverty and provide public goods, decisions have to be implemented to be effective. Successful implementation requires that the bureaucratic apparatus is competent and nonbiased and that the government carries out economic policies that are suitable to achieve its political goals.

Countries may be formally democratic even when several of these criteria are not satisfied. Deficiencies in the less formal aspects of governance may be the reason why the correlation between democracy and internal conflict is weak. If democratic institutions fail to produce good *informal* governance, they should be ineffective in reducing the risk of conflict. In the Formal versus Informal Institutions subsection, we will distinguish between the “formal” and “informal” institutions of governance and seek to separate their effects on the risk of internal armed conflict. We also look into the importance of socioeconomic development in the Institutions versus Average Income subsection. It is well known that formal democratic institutions are more frequently found and more stable in developed countries (e.g., Lipset 1959). The same applies to our informal institutions of governance (Fjelde and Hegre 2007). In the Institutions versus Average Income subsection, we look into whether the effect of governance on conflict can be distinguished from that of development.

When formal institutions place power with the majority of the population (as principals in a chain of representation), it is natural that this electorate instructs its delegates also to improve the quality of informal governance. Without it, after all, democratically based decisions will not be implemented. Formal democratic

institutions should lead to good informal institutions, then. In practice, however, it takes a long time for elected officials to substantially improve informal governance, and sometimes very little improvement is seen. Powerful actors may benefit from poor informal governance, and persistent poor governance also tends to erode the formal democratic institutions of a country. Fjelde and Hegre (2007) show that high levels of corruption increases the probability of change into semi-democracy or autocracy. In low-income or oil-dependent countries, this effect is stronger than the tendency for democratic institutions to reduce corruption. In high-income, economically diversified countries, democracies succeed in curbing corruption.

Measuring Governance

We have constructed seven indices of governance—six indices of informal institutions corresponding to the categories outlined previously, and one index of formal political institutions. Each of the informal institutions indices is constructed using two or more indicators. We also created a composite index of governance as an unweighted average of the seven subindices. All are normalized to have mean equal 0 and standard deviation 1, such that low values denote poor and high good governance. We perform multiple imputation (Little and Rubin 2002; Honaker and King 2010), which involves using all the information available in the data set to predict missing values as detailed in the Measuring Governance section. Descriptive statistics for our imputed data set are found in the Online Appendix, Table A-1.

Below, we summarize the indicators we have used. They are labeled by the institutions that created them: “ICRG” represents data from the International Country Risk Guide/Political Risk Group (PRS Group 2010); “WGI” from World Governance Indicators (Kaufmann, Kray, and Mastruzzi 2010, 4); “EF” from the “Economic Freedom Network” in collaboration with the Fraser Institute (Gwartney, Hall, and Lawson 2010, <http://www.freetheworld.com>);⁹ FH from Freedom House (Freedom House 2010); TI from Transparency International (<http://www.transparency.org>); WB from the World Bank (2009).

Bureaucratic Quality

ICRG Bureaucratic quality: High values are given to countries where “the bureaucracy has the strength and expertise to govern without drastic changes in policy or interruptions in government services” and “tends to be somewhat autonomous from political pressure and to have an established mechanism for recruitment and training.”

WGI Government effectiveness: Captures “perceptions of the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government’s commitment to such policies.”

The Rule of Law

WGI Rule of law: Captures “perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence.”

EF Area 2: covers “commercial and economic law and security of property rights.” It is an aggregate of indicators capturing the extent of military interference in rule of law and the political process, integrity of the legal system, regulatory restrictions on the sale of real property, and the legal enforcement of contracts.

FH Civil liberties: To be classified with top rating, countries must “enjoy a wide range of civil liberties, including freedom of expression, assembly, association, education, and religion. They have an established and generally fair system of the rule of law (including an independent judiciary), allow free economic activity, and tend to strive for equality of opportunity for everyone, including women and minority groups.”

Corruption

ICRG Corruption: The coding takes corruption in the form of demands for bribes into account, but “is more concerned with actual or potential corruption in the form of excessive patronage, nepotism, job reservations, ‘favor-for-favors,’ secret party funding, and suspiciously close ties between politics and business.”

WGI Control of corruption: Captures “perceptions of the extent to which public power is exercised for private gain, including both petty and grand forms of corruption, as well as ‘capture’ of the state by elites and private interests.”

TI Corruption Perception Index: An aggregate indicator which ranks countries in terms of the degree to which corruption is perceived to exist among public officials and politicians. Transparency International uses a wide range of sources comprising thirteen different surveys or assessments from several different institutions to construct the aggregate indicator.

Economic Policies

WB CPIA score: Assesses the extent to which a country’s policy and institutional framework supports sustainable growth, poverty reduction, and the effective use of development assistance. Scores from 2005 onward are publicly available (<http://data.worldbank.org/indicator>).¹⁰ Performance within four clusters are assessed: Economic Management, Structural Policies, Policies for Social Inclusion and Equity, and Public Sector Management and Institutions. Among the sixteen criteria are assessments of property rights and rule-based

governance, quality of budgetary and financial management, efficiency of revenue mobilization, quality of public administration, and transparency, accountability, and corruption in the public sector.

EF Area 1: Captures “Size of Government: Expenditures, Taxes, and Enterprises,” based on subindicators for general government consumption spending as a percentage of total consumption, transfers and subsidies as a percentage of GDP, government enterprises and investment, and top marginal tax rate.

EF Area 3: Covers “Access to Sound Money,” as indicated by money growth, the standard deviation of inflation, inflation in the most recent year, and citizens’ freedom to own foreign currency bank accounts.

EF Area 4: Covers “Freedom to Trade Internationally.” The index is based on indicators on taxes on international trade, black-market exchange rates, and capital controls.

EF Area 5: Captures “Regulation of Credit, Labour, and Business.” It builds on indicators of credit market regulations, labor market regulations, and business regulations.

EF Overall score: An aggregate of the five area scores.

Military in Politics

ICRG Military in politics: Assesses the military participation in government.

Military spending: Each country’s military spending as a share of GDP from the World Bank’s World Development Indicators (<http://data.worldbank.org/indicator>).

Political Exclusion and Repression

Political terror scale: Using country reports from the U.S. Department of State and Amnesty International, the Political Terror Scale classifies the repressive activities of states (Political Terror Scale [PTS]; Wood and Gibney 2010, <http://www.politicalterrorscale.org/ptsdata.php>).

Ethnic exclusion: Cederman, Wimmer, and Min (2010) have used country experts to systematically code several aspects of ethnic power relations (<http://www.icr.ethz.ch/research/epr>). The measure records the proportion of the population that are excluded based on their ethnic affiliation from decision-making authority within the central state power.

Formal Political Institutions

We have based our index of *formal political institutions* on the “Scalar Index of Politics” (SIP) developed by Gates et al. (2006). The SIP index is based on the Polity (Marshall n.d.) democracy index, the most widely used data set on formal political institutions in conflict research, and Polyarchy (Vanhanen 2000). See the Online Appendix for more details on how we use this index.

Governance and Conflict

In this section, we study how our seven disaggregated indicators of governance and the aggregated governance indicator relate to the risk of conflict onset and recurrence. In the next section, we decompose the effect of governance into what can be attributed to formal democratic institutions as distinct from informal aspects of governance. Finally, we seek to distinguish between governance on one hand and economic development on the other.

Design

We have chosen a design that allows us to analyze how our governance indicators relate to the *incidence* of civil war. Our data set includes all independent countries for which data exist for all years from 1960 through 2008. The dependent variable is whether an internal armed conflict is going on in a country year, as coded in the Uppsala Conflict Data Program/Peace Research Institute Oslo (UCDP/PRIO) data set (Gleditsch et al. 2002; Harbom and Wallensteen 2010). We include all conflicts with at least twenty-five battle-related deaths per year, restricting attention to countries listed as the *location* of conflict in the UCDP/PRIO data set. All models include a two-category lagged independent variable—the estimates for the other variables should therefore be interpreted as the extent to which the variables change log odds of conflict incidence given the conflict state the year before.

To facilitate a special focus on conflict recurrence, the models also include a term called “peace years”—the log of the number of years without conflict up to $t - 1$. If the country had a conflict at $t - 1$, this variable is set to zero. The estimate for the peace years variable captures how the risk of conflict changes with time in peace given that there was no conflict in the previous year, which is roughly the same as the risk of conflict recurrence.¹¹ To separate how governance affects the risk of conflict onset from how it reduces the risk of recurrence, we have entered an interaction term between the governance indicators and the peace years variable.

We control for whether a neighboring country has conflict, log GDP per capita, whether one ethnic group is demographically dominant, log population size, and the presence of United Nations peace keeping operations (PKOs).¹² In addition, we have included twenty-three region dummies to partly account for unobserved regional heterogeneity. The definition and justification of the regions is found in the Online Appendix, Section A-5.

Results, Indicators of Governance

Table 1 reports the results from an empirical analysis of how our governance variables affect the risk of conflict recurrence. Models 1 to 7 estimate the effect of each of our seven governance indicators, and model 8 that of our combined governance

Table 1. Internal Conflict as Function of Eight Governance Indicators, 1960 to 2008, Logit Regression.

	(1) Bureaucracy	(2) Rule of law	(3) Corruption	(4) Economics	(5) Military	(6) Exclusion	(7) Formal	(8) Governance
Conflict								
Conflict, $t - 1$	2.979*** (0.178)	2.993*** (0.178)	2.990*** (0.178)	2.976*** (0.178)	2.987*** (0.179)	2.861*** (0.180)	2.967*** (0.179)	2.992*** (0.178)
War, $t - 1$	4.313*** (0.295)	4.292*** (0.298)	4.267*** (0.293)	4.240*** (0.298)	4.221*** (0.294)	3.928*** (0.300)	4.267*** (0.294)	4.287*** (0.298)
ln(Peace years)	-0.660*** (0.0697)	-0.645*** (0.0696)	-0.664*** (0.0706)	-0.632*** (0.0687)	-0.611*** (0.0685)	-0.590*** (0.0683)	-0.617*** (0.0680)	-0.650*** (0.0698)
Bureaucratic quality index	0.198 (0.121)							
Bureaucratic quality $\times$ peace years	-0.217*** (0.0577)							
Rule of law index		0.104 (0.130)						
Rule of law $\times$ peace years		-0.169** (0.0598)						
Corruption index			0.193 (0.125)					
Corruption $\times$ peace years			-0.212*** (0.0605)					
Economic policies index				0.0111 (0.0990)				
Economic policies $\times$ peace years				-0.133*** (0.0505)				
Military influence index					-0.0235 (0.100)			
Military influence $\times$ peace years					-0.108* (0.0502)			
Pol. exclusion and rep. Index						-0.300** (0.0951)		
Pol. excl. $\times$ peace years						0.0479 (0.0514)		

	Formal inst. index	Formal inst. $\times$ peace years	Governance index, $t-1$	Governance index $\times$ peace years	Log population	log GDP per capita, $t-1$	PKO, traditional	PKO, transformational	Neighbor conflict, $t-1$	Ethnic dominance	Intercept	AIC	LL	N
	0.378*** (0.0541)	0.375*** (0.0536)	0.371*** (0.0535)	0.393*** (0.0541)	0.385*** (0.0536)	0.358*** (0.0551)	0.378*** (0.0537)	0.374*** (0.0536)	0.341** (0.153)	0.380* (0.150)	3.717*** (0.946)	2.267.7	-1.099.3	6,698
	-0.129 (0.120)	-0.128 (0.115)	-0.145 (0.114)	-0.101 (0.112)	-0.134 (0.106)	-0.166 (0.104)	-0.254* (0.107)	-0.125 (0.117)	-0.0244 (0.230)	-4.309*** (0.914)	2.267.7	2.266.7	-1.095.3	6,698
	-0.00279 (0.227)	0.0322 (0.226)	0.00346 (0.227)	0.0415 (0.226)	0.0192 (0.227)	-0.0240 (0.228)	-0.0244 (0.230)	0.0200 (0.226)	-0.891 (0.476)	0.364* (0.150)	2.267.7	2.266.7	-1.095.3	6,698
	-0.660 (0.475)	-0.709 (0.476)	-0.683 (0.474)	-0.745 (0.477)	-0.795 (0.479)	-0.942 (0.482)	-0.891 (0.476)	-0.714 (0.475)		-4.309*** (0.914)	2.267.7	2.266.7	-1.095.3	6,698
	0.362* (0.155)	0.347* (0.154)	0.362* (0.153)	0.335* (0.153)	0.347* (0.152)	0.289 (0.153)	0.341* (0.153)	0.344* (0.154)		0.364* (0.150)	2.267.7	2.266.7	-1.095.3	6,698
	0.336* (0.150)	0.347* (0.151)	0.336* (0.151)	0.369* (0.149)	0.372* (0.149)	0.364* (0.150)	0.380* (0.150)	0.352* (0.150)		0.364* (0.150)	2.267.7	2.266.7	-1.095.3	6,698
	-4.636*** (1.099)	-4.659*** (1.030)	-4.451*** (1.014)	-5.032*** (1.014)	-4.815*** (0.945)	-4.309*** (0.914)	-3.717*** (0.946)	-4.665*** (1.038)		-4.309*** (0.914)	2.267.7	2.266.7	-1.095.3	6,698
	2.260.9	2.266.7	2.262.4	2.265.9	2.267.8	2.264.8	2.267.7	2.266.7		2.267.7	2.266.7	2.266.7	-1.095.3	6,698
	-1.096.4	-1.099.4	-1.097.2	-1.099.0	-1.099.9	-1.098.4	-1.099.8	-1.095.3		-1.099.8	-1.099.8	-1.099.8	-1.095.3	6,698
	6.698	6.698	6.698	6.698	6.698	6.698	6.698	6.698		6.698	6.698	6.698	-1.095.3	6,698

Note: Standard errors in parentheses. Estimation also includes twenty-three-regional dummies, results not shown. AIC = akaike information criterion. LL = log likelihood. PKO = peace keeping operations. [†] $p < .05$, [‡] $p < .01$, ^{***} $p < .001$.

measure. The results show that in general, the probability that a country is in conflict is much higher if it was at conflict last year—the Conflict and War at $t - 1$ terms are large, positive, and clearly significant. Likewise, the risk of conflict recurrence gradually diminishes for each new consecutive year of peace. The $\ln(\text{Peace years})$ terms are negative and significant, ranging from -0.59 to -0.66 in size. They indicate the change in log odds of recurrence when the duration of peace is multiplied by 2.7. The odds of conflict is reduced by 45 to 50 percent if peace holds for the first 2.7 years. The estimated risk is reduced by the same factor if peace holds up to year 7, and once again if it holds up to year 20.¹³

The main terms for six of our eight governance indicators are not statistically different from zero—good governance has a negligible effect on the risk of conflict when time in peace is close to zero. Seven of the interaction terms, on the other hand, are negative and significant. With good governance, the risk of conflict recurrence is reduced more quickly. The seven interaction term estimates range from -0.099 to $-.219$. Compared to a country with poorer governance, a country with a one standard deviation higher governance score sees an additional risk reduction of 10 to 20 percent over the first 2.7 years.

To illustrate these effects, Figure 1 shows the predicted probability of conflict as a function of time since last conflict or independence for four of the models—for bureaucratic quality, exclusion and repression, formal institutions, and the combined governance index.¹⁴ The plot shows the risk for two hypothetical countries that are similar except that one has poor governance (governance index = -1.28 or about the 10th percentile) and the other good governance ($1.28/90$ th percentile). The values for other predictors are set at the global averages in 2005.

Systems with high bureaucratic quality may have slightly higher risk of conflict immediately after a conflict or independence. As time goes by, however, the risk decreases much more rapidly when bureaucratic quality is good than when it is poor. The conflict risk is twice as high in low-governance systems after six to eight years, and three times higher after twenty years. The plots for rule of law, corruption, and economic policies are very similar to this plot (see the Online Appendix, Figure A-4), as does that for our combined governance indicator. The plot for the formal institutions index shows another pattern. For this indicator, well-governed countries seem to have a higher initial risk of conflict. This difference is steadily reduced over time, and changes sign after about eighteen years. After another couple of decades, well-governed (democratic) countries have a lower risk of conflict than nondemocratic ones. Only the good governance seen as absence of exclusion or repression seems to reduce the risk of conflict onset.

The fact that most governance indicators pull in the same direction is useful for two reasons. First, it allows us to treat “governance” as a unified concept that is efficiently captured by our combined governance indicator. We limit our discussion to this indicator in the remainder of the article. Second, and more important, these results mean that all six aspects of informal governance covered here are important to reduce the risk of conflict recurrence, implying that different forms of governance

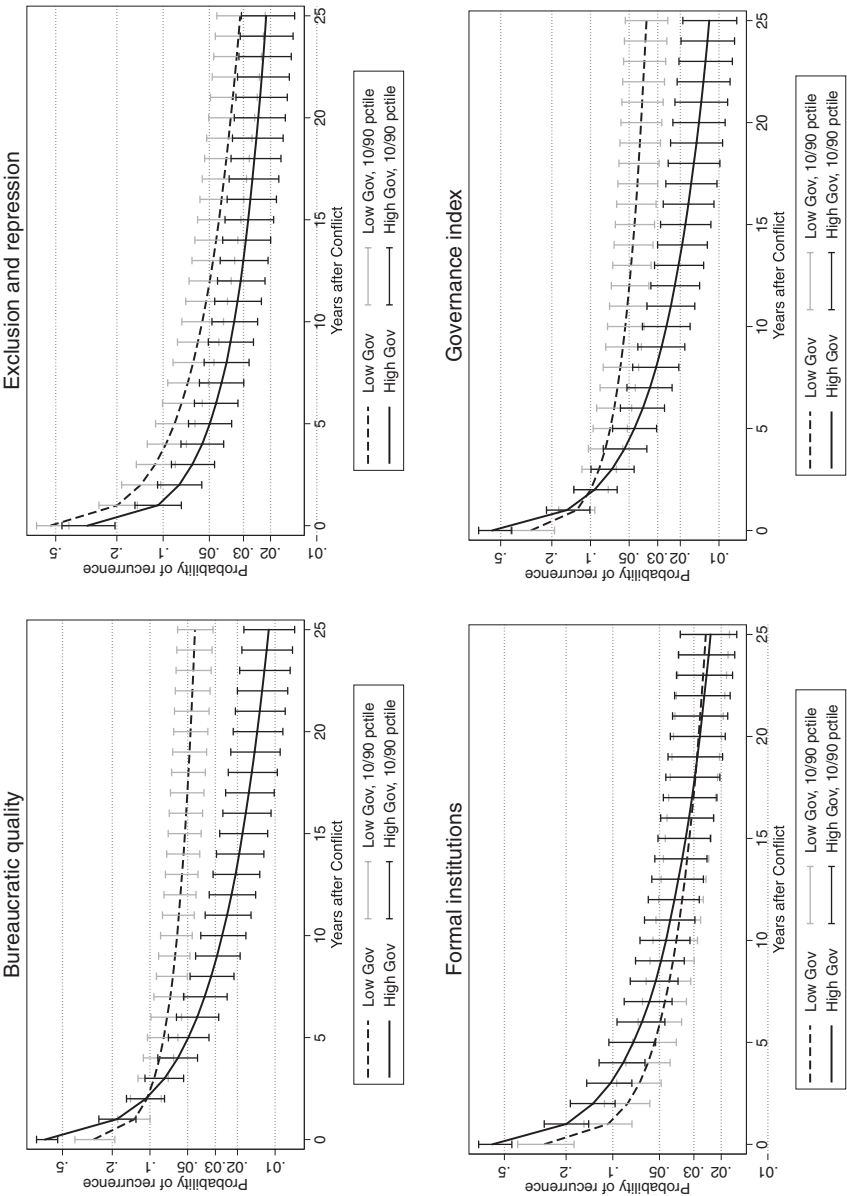


Figure 1. Predicted probability of conflict recurrence, governance indicators.

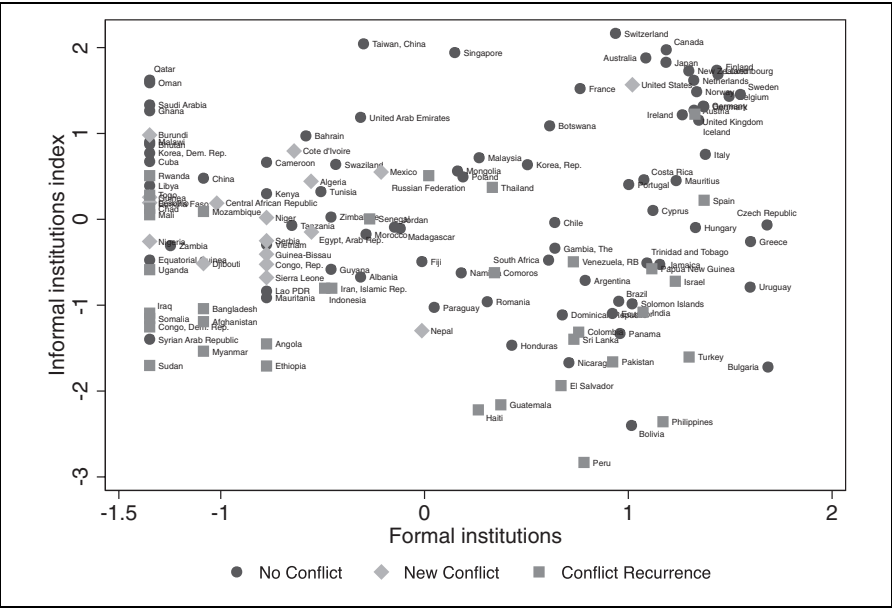


Figure 2. Informal institutions by formal institutions, 1990.

are partial substitutes to each other. The absence of an aspect of good governance may be offset by the presence of another.

Formal versus Informal Institutions

It is clear from Table 1 that good governance reduces the risk of conflict recurrence. But to what extent is the effect of our governance indices distinct from that of formal (democratic) institutions, or from the economic resources available to governments?

To explore the first question, we created an index of “informal political institutions” which is simply the residuals from the regression line formed by the formal institutions index as x variable and the governance index as y variable.¹⁵ The formal institutions index explains 65 percent of the variance in the governance index, and a one-unit increase in formal institutions means, on average, an increase of 0.8 in overall governance.

The residuals from this model were rescaled to have a standard deviation of one, forming our “informal institutions index.” Figure 2 shows the relationship between the formal (x axis) and the informal (residual) institutions (y axis) for the year 1990. Some countries, such as Oman, United Arab Emirates, Singapore, Switzerland, and Canada do much better in terms of governance than indicated by their formal institutions—the “informal institutions” residual index is high. Others, such as Sudan, Ethiopia, Haiti, and Peru do much worse.

The conflict experiences of the countries are also depicted in Figure 2. Countries with no conflicts over the 1991–2005 period are marked with filled circles. Countries that had no conflict during 1946 to 1990 but a new conflict during the 1991 to 2005 period are marked with diamonds. Countries with pre-1990 conflicts recurring after 1990 are marked with a square. As expected from Table 1, it is clear from Figure 2 that countries with poor governance have had more frequent conflicts and conflict recurrences. No-conflict countries are concentrated in the right half of the figure, and particularly in the upper right corner. Countries such as Botswana, Costa Rica, and Sweden have good formal and informal governance and no conflict. The exceptions to the pattern of conflict and good governance are the conflicts with the Euzkadi Ta Askatasuna (ETA), Irish Republican Army (IRA), and Al-Qaida in Spain, United Kingdom, and the United States, respectively.¹⁶

The new conflicts (diamond markers) are primarily found in the left half of Figure 2, in countries with very incomplete democratic institutions in 1990, but often with fairly good informal governance. Examples are Burundi, Cote d'Ivoire, and Mexico.

Conflict recurrences after 1990 (squared markers) are concentrated in the lower half of the figure, that is, in countries with poor informal governance relative to their formal governance. Examples are Sudan, Ethiopia, and Haiti among the nondemocracies, and Turkey and the Philippines among the more formally democratic systems. Most exceptions to this pattern of poor governance and conflict are countries with fairly robust formal institutions, such as Bolivia or Bulgaria—these, after all, are well-governed overall since their formal institutions are democratic. One other exception is Syria, which turned out to be less exceptional after the end year of our data set (2008).

Table 2 shows the results of estimating a model including both the formal institutions index, the informal institutions index, as well as their interaction terms with the “peace years” variable. Model 9 (column 1) includes only the formal institutions index variables. The results are similar to model 7, Table 1. In model 10, the formal institutions variables are replaced by the informal (residual) index and interaction term. The interaction term with peace years is negative and about the same size as for the formal institutions index. The main term, on other hand, is negative and close to statistical significance. Good informal institutions relative to formal institutions reduces the risk of conflict also in the early years after independence or earlier conflicts. The Akaike Information Criterion (AIC) of model 10 is clearly lower than for model 9.

In model 11, we include all four terms to assess the relative importance of formal and informal terms. Only the interaction terms are significant, but the log likelihood of the model is considerably higher than for the two preceding models, and the AIC is lower. As in the other two models, the formal institution's main term is positive and the informal institution's main term negative. Neither are clearly different from zero, but the difference between them is significant.

Figure 3 indicates how the results of model 11 should be interpreted. The left panel shows the median predicted probability of conflict just after conflict or

Table 2. Internal Conflict as Function of Formal and Informal Institutions of Governance, GDP Per Capita, "Surplus Governance," 1960 to 2008, Logit Regression and Instrumental Analysis.

	(9) Formal only GDP/cap	(10) Informal only GDP/cap	(11) Formal informal GDP/cap	(12) GDP/cap only	(13) GDP/cap surplus Gov	(14) GDP/cap surplus formal	(15) Governance Index	(16) Conflict
Main								
Conflict, $t - 1$	2.966*** (0.179)	2.959*** (0.178)	2.973*** (0.179)	2.951*** (0.185)	2.971*** (0.186)	2.956*** (0.186)	1.635*** (0.145)	0.0520 (0.0400)
War, $t - 1$	4.266*** (0.294)	4.153*** (0.295)	4.231*** (0.300)	4.231*** (0.299)	4.282*** (0.304)	4.227*** (0.305)	2.419*** (0.219)	-0.251*** (0.0491)
ln(Peace years)	-0.617*** (0.0680)	-0.597*** (0.0694)	-0.617*** (0.0710)	0.303 (0.368)	0.498 (0.391)	0.635 (0.543)	-0.351*** (0.0519)	0.0787*** (0.0108)
ln(GDP per capita), $t - 1$	-0.252* (0.107)	-0.0560 (0.107)	-0.0679 (0.118)	-0.0548 (0.121)	-0.0321 (0.123)	-0.242 (0.153)	-0.289 (0.197)	0.455*** (0.0158)
GDP per capita $\times$ peace years				-0.118* (0.0467)	-0.145*** (0.0501)	-0.157*** (0.0695)		
Surplus governance index					0.0743 (0.0989)	-0.184 (0.140)		
SGI $\times$ peace years					-0.103* (0.0513)	-0.130 (0.0733)		
Formal inst. index	0.292* (0.103)		0.159 (0.112)			0.410* (0.161)		
Formal inst $\times$ peace years	-0.0986 (0.0504)		-0.120* (0.0527)			0.0368 (0.0843)		
Log population	0.378*** (0.0537)	0.379*** (0.0537)	0.366*** (0.0540)	0.373*** (0.0547)	0.363*** (0.0549)	0.354*** (0.0553)	0.158*** (0.0413)	-0.00459 (0.00646)
Neighbor conflict, $t - 1$	0.341* (0.153)	0.294 (0.153)	0.305* (0.155)	0.332* (0.156)	0.331* (0.157)	0.294 (0.158)	0.184 (0.120)	-0.0108 (0.0203)
Ethnic dominance	0.384* (0.150)	0.368* (0.150)	0.348* (0.151)	0.358* (0.152)	0.350* (0.153)	0.342* (0.154)	0.200 (0.111)	-0.0146 (0.0188)
PKO, traditional	-0.0355 (0.229)	-0.0323 (0.226)	-0.0872 (0.229)	0.00588 (0.230)	-0.0208 (0.230)	-0.124 (0.233)	-0.0217 (0.169)	-0.0190 (0.0391)
PKO, transformational	-0.841 (0.478)	-0.888 (0.482)	-0.847 (0.479)	-0.661 (0.479)	-0.632 (0.477)	-0.813 (0.481)	-0.559 (0.448)	0.169 (0.113)

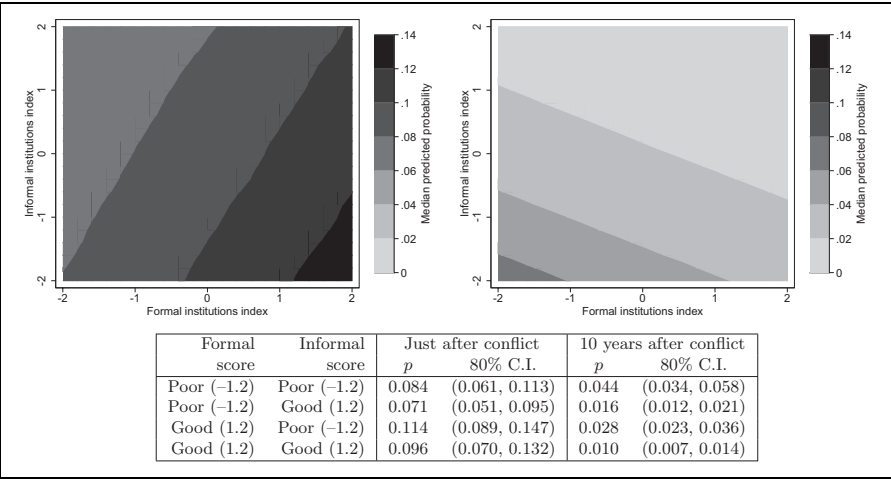


Figure 3. Predicted probability of conflict by extent of formal and informal governance, just after conflict (left panel) and ten years after conflict (right panel), based on model 11.
Note: Uncertainty estimates for selected points reported subsequently.

independence, assuming typical values for control variables.¹⁷ The predicted risk immediately after conflict or independence is highest in the lower right corner—in countries with good formal institutions but poor informal institutions (e.g., the Philippines or Peru), and decreasing as the quality of the informal institutions improve. The right panel shows the predicted probability 10 years after conflict. At this point, the risk is highest in countries that score low on both indices (as in Myanmar and Sudan), and decrease with improvement along both governance dimensions.

These predictions come with considerable uncertainty. Below the plots, we report the predicted probability of conflict with confidence intervals for some combinations of values. Just after the conflict, the median probability of conflict (recurrence) for a country with scores of -1.2 (about the 20th percentile) for both indices is 0.084. The 80 percent confidence interval for this prediction is (0.061, 0.113). The estimated uncertainty is considerable. It is fairly clear, however, that countries with nondemocratic institutions but excellent informal governance have lower immediate risk of recurrence than states with democratic institutions but poor informal governance. The reduction in risk is strongest for countries with good governance along both dimensions. A country that scores well on both dimensions has a predicted risk four times lower than a similar country that scores poorly on both.

To summarize, both the formal and informal institutions of governance must be good in order to effectively reduce the risk of conflict. However, the informal institutions seem to be even more important substantially than the formal or democratic institutions.

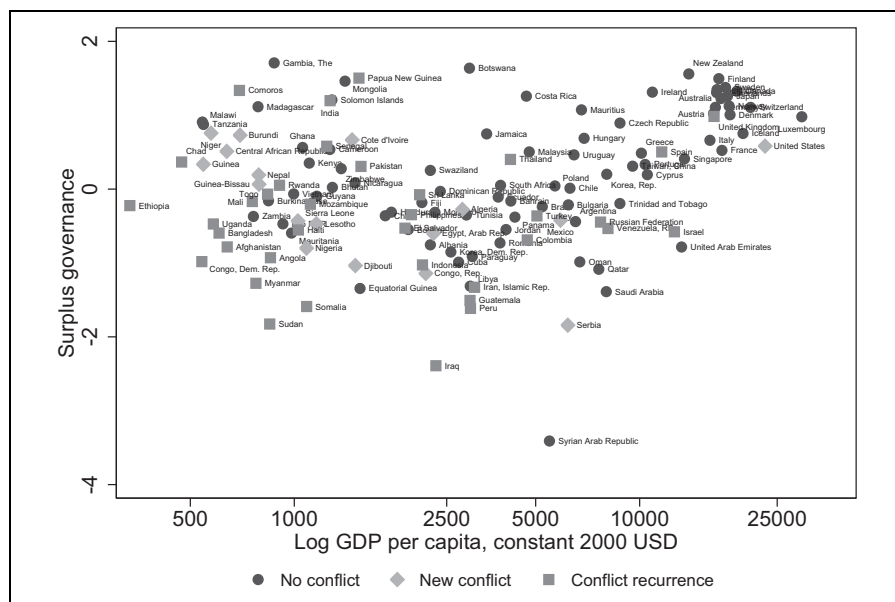


Figure 4. “Surplus governance” by average income, 1990.

Note: “Surplus governance” is defined as the residual of regressing governance on log gross domestic product (GDP) per capita, normalized to range from 0 to 1.

Institutions versus Average Income

An inspection of Figure 2 shows that countries that score high on both the governance and the formal institutions index are relatively rich countries (e.g., Sweden, Switzerland, Ireland). Most countries with low scores for the two indicators are poor (Sudan, Angola, and Bangladesh). We demonstrated in Table 1 that good governance reduces the risk of conflict controlling for poverty (as measured by GDP per capita). GDP per capita is among the most robust variables in the armed conflict literature (Hegre and Sambanis 2006). The estimate for the variable is negative in all our models, but significantly different from zero in only one. This may imply that good governance is an important intermediate variable between income and domestic peace. To look more closely into how wealth and governance interrelate, we have defined a “surplus governance” index as the residual from regressing our combined governance index on log GDP per capita.¹⁸

Figure 4 plots these “surplus governance” residuals against GDP per capita for the year 1990. Some countries are considerably better governed than expected from their average income levels. That India, Botswana, and New Zealand are relatively well governed is well known. In addition, Malawi, Mongolia, and Costa Rica have surprisingly high values for our governance index. Sudan, Peru, Serbia (Yugoslavia), and Syria, on the other hand, turn out as relatively poorly governed in 1990. A set of heavily oil-

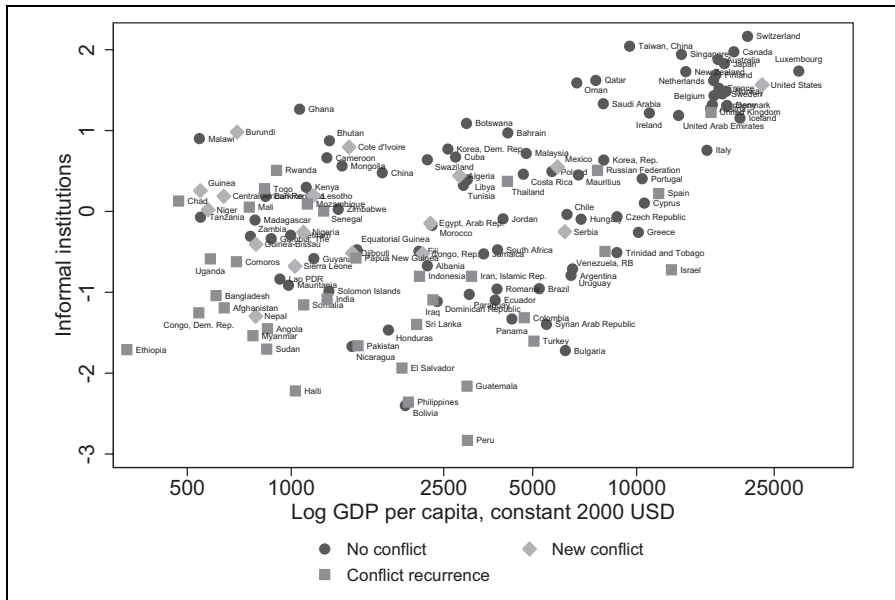


Figure 5. “Informal governance” by average income, 1990.

Note: “Informal governance” is defined as the residual of regressing governance on our formal democratic institutions.

dependent countries (e.g., Qatar, Saudi Arabia, United Arab Emirates) also stand out as poorly governed relative to their average incomes, as argued by Fearon (2005).

As is well known from previous studies, Figure 4 shows that most conflict and conflict recurrence countries (marked with diamonds and squares) have low average incomes. Spain, the United Kingdom, and the United States are the main exceptions. Although patterns are less clear than in Figure 2, there is a tendency for conflicts to recur more frequently in low-income countries that had relatively poor governance for their income levels in 1990. Examples are DR Congo, Sudan, and Guatemala, with Syria as the most evident exception. New conflicts, again, seem to occur more frequently in relatively well-governed but poor countries.

To some extent, the apparent poor governance of these countries is due to lack of formal democratic institutions. In Figure 5, we have plotted the residual “informal governance” index against average income. Seen from this perspective, oil exporters such as Oman, Saudi Arabia, and the United Arab Emirates appear as well-governed nondemocracies, whereas fairly democratic countries such as Venezuela, Lebanon, and Greece have poor informal governance relative to their income levels. As before, there is a tendency for nonconflict countries to be in the upper, well-governed half of the figure.

Given this correlation, to what extent does governance explain the risk of conflict recurrence beyond what can be attributed to a country’s average income level? Table 2

reports results from models including our “surplus governance” indicator. Model 12 (column 4) includes only GDP per capita and its interaction with “peace years.” The main term is not significant on its own, but the interaction with time in peace is. This finding may be less controversial than it seems, since our model is a model of conflict incidence, controlling for whether the country was at conflict at $t - 1$.

Model 13 includes the “surplus governance” indicators. As found for the other governance indicators, “surplus governance” has a negative relationship with the risk of conflict recurrence—the estimates are roughly similar to those depicted in Figure 1 based on model 8, Table 1.

Figure 6 helps interpreting the results in model 13. Immediately after conflict (the left panel), the estimated risk of renewed conflict is about 0.10 annually for all types of countries: neither the GDP per capita nor “surplus governance” are able to distinguish clearly between high- and low-probability countries. Ten years after, however, both surplus governance and average income have made a big difference. Poor countries with poor governance for their income levels have not decreased their expected risk of conflict recurrence, whereas countries that are either well governed for their income levels or have high average incomes have reduced their risks to about a quarter of the initial level. The lowest estimated risk is, unsurprisingly, in countries with high income and good governance. According to our Clarify simulations, the well-governed cases have a lower risk of conflict than do poorly governed ones, for both income levels.

Endogeneity

It is not obvious that conflict and governance are exogenous to each other. Armed conflict often leads governments to introduce martial law and limit political rights, the opposition sometimes to boycott elections, and all parties to increase violence connected to elections. The poor governance of Sudan and Angola in 1990 might very well be due to their past conflict history.

To test for endogeneity bias, we estimate an instrumental variable model based on model 8 in Table 1. The results are shown in the last two columns of Table 2. There are two key explanatory variables in the original model: the country’s governance level and the governance level interacted with time in peace. As instrumental variables for governance, we use (1) a measure of democratic waves as proposed by Knutsen (2011), based on the data in Dahl et al. (2013) and (2) the twenty-year lag of the formal institutions index (Helliwell 1994). Since models with more than one instrumented variable are difficult to estimate by means of maximum likelihood estimation (MLE), we use a two-step model proposed by Newey (1987). This model is consistent, but less efficient than the MLE approach.

Model 15 gives the results from the first stage of the instrumental variable regression, model 16 those of the second stage. The substantive effects are largely similar to the effects seen in Table 1. The size of the effects are somewhat different, reflecting that the instrument is an imperfect rendition of the original variable. The estimation

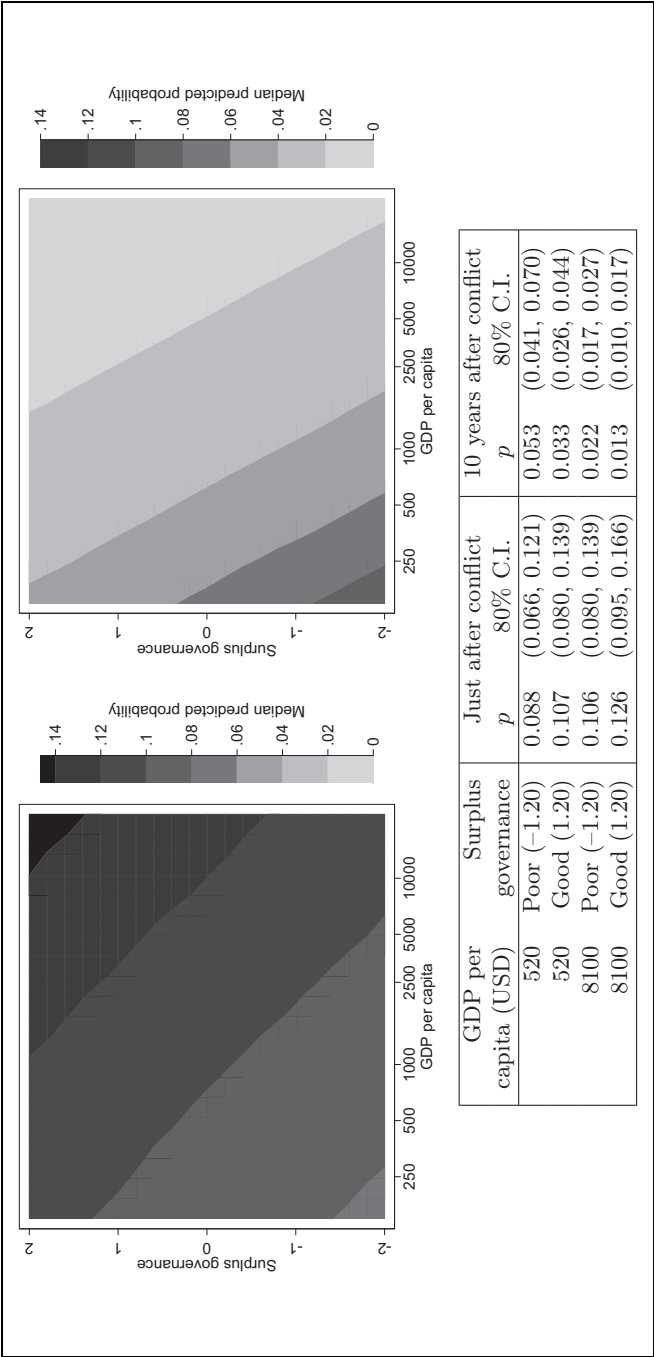


Figure 6. Predicted probability of conflict by extent of gross domestic product (GDP) per capita and “surplus” governance, just after conflict (left panel), and ten years after conflict (right panel), based on model 13.
Note: Uncertainty estimates for selected points reported subsequently.

shows that the instruments we use do not violate the exclusion criteria. It also shows that the estimated effect of governance instrumented is roughly the same as seen for the possibly endogenous variable in Table 1. In sum, we conclude that the endogeneity bias in the models shown in Table 1 is moderate and prefer to relate to the more efficient original estimates.

Conclusion

This article has investigated whether well-governed countries are better able to prevent armed conflict than poorly governed ones, with a particular focus on avoiding conflict recurrence. We have defined governance more broadly than in earlier studies, supplementing indices of “formal” degree of democraticness with less formal dimensions of governance. This broad conception of governance better captures the logic of explanations for why some political systems should have less conflict, such as the relative-deprivation account associated with Davies (1962) and Gurr (1968). Countries may be formally democratic but have poor governance according to this definition. Deficiencies in informal aspects of governance may be the reason why the correlation between democracy and internal conflict is weak (Hegre et al. 2001)—poor informal governance may completely undermine the decisions reached in democratic institutions, and good governance may reduce deprivation effectively even in nondemocratic systems.

We have studied a set of indicators classified into seven aspects of governance: bureaucratic quality, the rule of law, corruption, economic policies, military influence in politics, political exclusion and repression, and formal political institutions, and operationalized these indices by means of data from a wide range of available sources. Our statistical models show that good governance is crucial to reducing the risk of conflict recurrence. Countries that have experienced conflict have a higher risk of seeing renewed conflict. The risk of renewed conflict in countries with good governance, however, drops rapidly after the conflict has ended. In countries characterized by poor governance, this process takes much longer.

At the same time, there are good reasons to believe that conflicts often erodes the quality of governance. To account for this potential endogeneity, we constructed an instrument for governance that is highly correlated with the observed governance indicator and yet clearly exogenous to conflict. We reach the same conclusions using the instrumented governance variable.

We have also studied the importance of *formal* political institutions (the extent to which countries are regarded as democratic or not) relative to *informal* institutions. The results indicate that countries need good institutions along both dimensions to minimize the risk of internal armed conflict, but that informal governance might be somewhat more important than formal institutions.

Moreover, we have shown that good formal and informal governance is a partial explanation of the conflict-reducing effect of economic development. High-income countries are on average well governed. Still, a model that includes both GDP per

capita and governance fits the data better than a model with only GDP per capita. We find a discernible difference in the risk of conflict recurrence between high-income countries that are well governed and countries that have poor governance relative to their income levels.

Even though a combination of good formal and informal institutions is optimal, the analysis also shows that all our governance indicators are related to a decreased risk of conflict recurrence. We take this as indication that reform in the different sectors can be partial substitutes to each other. Any reform that improves governance may be conflict-reducing, be it to the formal political institutions, bureaucratic quality, or corruption.

Our analysis sheds light on the somewhat inconclusive results from earlier studies of the relationship between democracy and conflict, which find a heightened risk of conflict in semi-democracies (Hegre et al. 2001; Fearon and Laitin 2003). It is possible that some of the conflict-proneness of semi-democracies is due to their poor governance in a wider sense. Given that good governance as defined here is more frequent in middle- and high-income countries, our results are also in line with Hegre (2003) and Collier and Rohner (2008) who find democracy to reduce the risk of conflict more strongly the higher is GDP per capita.

In substantial terms, the difference between high-income and low-income countries is much larger than the effect produced by different levels of governance. However, even with excellent economic policies (one aspect of good governance), it takes many decades to close the gap between a low-income and a high-income country. Good informal governance is a much more expedient means to reduce the risk of conflict in the long run. In the longer run, strong economic growth and democratic institutions will reinforce the beneficial effects of informal governance.

Authors' Note

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Declaration of Conflicting Interests

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Supplemental Material

The online appendices are available at jcr.sagepub.com/supplemental. The online appendix and replication data are also posted at havardegre.net/items/governance.

Notes

1. As Råby and Teorell (2010) argues for the case of interstate conflict.
2. Fearon (2011) treat some of the same governance indicators in an analysis of conflict onset.
3. Even increasing inequalities may be perceived as acceptable. For instance, the rising economic inland-coastal inequalities in China over the last twenty years (Kanbur and Zhang 2005, 96) is probably perceived to be due to the exports-driven growth at the coast rather than deliberate policies to hold the inland back. Migration, rather than rebellion, is the natural response to such developments.
4. Some studies do find a relationship between intergroup inequality and a high risk of armed conflict (Østby 2008; Cederman, Weidmann, and Gleditsch 2011). In such cases, it is more natural that a government is seen as exacerbating such differences through their policies of exclusion and discrimination.
5. A bureaucracy that is able to implement decisions with great efficiency may prove extremely lethal when governments deliberately use them in violent campaigns, such as in Stalin's Union of Soviet Socialist Republics (USSR) or Hitler's Germany. Our definition may partly regard such bureaucracies as aspects of "good governance." However, merit-based recruitment, hardly the main characteristic of these organizations, is also important for our definition. As it is, there are few similar cases in our 1960–2008 period of study.
6. In a recent review and assessment of state capacity measures, Hendrix (2010) lists twelve different operationalizations of the concept. Some of these operationalizations are directly linked to state capacity, while others tend to conflate characteristics of regime type with the regime's capabilities. All these measures are correlated and closely related to our other indicators of governance.
7. The issues treated in this section draws in part on Hegre and Fjelde (2009), which treats the questions related to postconflict democracy and conflict recurrence in much more detail.
8. For a comprehensive review of the relationship between democracy and conflict, see Hegre (2014).
9. Economic freedom is defined as "when (a) property [individuals] acquire without the use of force, fraud, or theft is protected from physical invasions by others and (b) they are free to use, exchange, or give their property as long as their actions do not violate the identical rights of others" (Ben Nasser Al Ismaily et al. 2010, 3).

10. We have access to older scores but are not allowed to disclose them except as aggregated over regions.
11. We do not distinguish between periods of peace that started with the termination of a conflict and those that started with the independence of a country. For simplicity, we will often refer to conflict after previous conflict or independence as “recurrence.” Merging these two situations is obviously a simplification. However, the immediate postindependence and postconflict periods share some characteristics. In both cases, political institutions have been extensively reformed and new actors have come to power. Often, independence was achieved after a militarized contest. About half of the onsets in our data set came after previous (observed) conflict, about half after independence. Forty-two percent of the countries in the data set never had conflict.
12. The gross domestic product (GDP) per capita variable is calculated based on data from Maddison (2007), World Bank (2011), and Gleditsch (2002), measured in international Geary-Khamis dollars (Int\$), and log-transformed. The ethnic dominance variable is from Collier and Hoeffler (2004) and is coded 1 if one ethnic group in the country comprises a majority of the population. The population variable originate from the World Population Prospects 2006 (United Nations 2007), and the peace keeping operation (PKO) variables are from Hegre, Hultman, and Nygård (2011).
13. In the Online Appendix, Figure A-5, we show that the relationship between time in peace and conflict is roughly linear.
14. The predictions were obtained by means of the Clarify package (King, Tomz, and Wittenberg 2000).
15. The relationship between the governance and formal institutions indices is shown for the year 1990 in the Online Appendix, Figure A-3.
16. The UCDP project codes the attack of September 11, 2001, as an internal armed conflict in the United States.
17. We used Clarify (King, Tomz, and Wittenberg 2000) to obtain these predictions averaged across 1,000 simulations. We set control variable to typical values—log GDP per capita to 8.02 or Int\$ 3040, log population to 8.87 or 7.1 million, population growth to 1.9 percent, ethnic dominance to .50, and other variables to zero in our Clarify simulations.
18. Fearon (2011, p. 45) also makes use of a similarly constructed index of ‘surprisingly good governance’.

References

- Ben Nasser Al Ismaily, Salem, Fred McMahon, Miguel Cervantes, and Amela Karabegovic. 2010. *Economic Freedom of the Arab World: 2010 Annual Report*. Fraser Institute. Accessed January 16, 2014. <http://www.freetheworld.com>.
- Braithwaite, Alex. 2010. “Resisting Infection: How State Capacity Conditions Conflict Contagion.” *Journal of Peace Research* 47 (3): 311-19.
- Bussmann, Margit, and Gerald Schneider. 2007. “When Globalization Discontent Turns Violent: Foreign Economic Liberalization and Internal War.” *International Studies Quarterly* 51 (1): 79-97.

- Carey, Sabine C. 2007. "Rebellion in Africa: Disaggregating the Effect of Political Regimes." *Journal of Peace Research* 44 (1): 47-64.
- Cederman, Lars-Erik, Simon Hug, and Lutz F. Krebs. 2010. "Democratization and Civil War: Empirical Evidence." *Journal of Peace Research* 47 (4): 377-94.
- Cederman, Lars-Erik, Nils B. Weidmann, and Kristian S. Gleditsch. 2011. "Horizontal Inequalities and Eth-nonationalist Civil War: A Global Comparison." *American Political Science Review* 105 (3): 478-95.
- Cederman, Lars-Erik, Andreas Wimmer, and Brian Min. 2010. "Why Do Ethnic Groups Rebel? New Data and Analysis." *World Politics* 62 (1): 87-119.
- Collier, Paul, and Anke Hoeffler. 2004. "Greed and Grievance in Civil War." *Oxford Economic Papers* 56 (4): 563-95.
- Collier, Paul, and Dominic Rohner. 2008. "Democracy, Development and Conflict." *Journal of the European Economic Association* 6 (1): 531-40.
- Dahl, Marianne, Scott Gates, Håvard Hegre, and Håvard Strand. 2013. "Why Waves? Global Patterns of Democratization, 1820–2008." Typescript, PRIO. Accessed on February 2, 2014. <http://havardhegre.net/items/why-waves/>
- Davenport, Christian. 2007. "State Repression and Political Order." *Annual Review of Political Science* 10:1-23.
- Davies, James C. 1962. "Towards a Theory of Revolution." *American Sociological Review* 27 (1): 5-19.
- Dreher, Axel, Martin Gassebner, and Lars-H. R. Siemers. 2012. "Globalization, Economic Freedom, and Human Rights." *Journal of Conflict Resolution* 56 (3): 516-46.
- ESCWA. 2011. "The Governance Deficit and Conflict Relapse in the ESCWA Region". New York: The UN.
- Fearon, James D. 2004. "Why Do Some Civil Wars Last So Much Longer Than Others?" *Journal of Peace Research* 41 (3): 275-301.
- Fearon, James D. 2005. "Primary Commodity Export and Civil War." *Journal of Conflict Resolution* 49 (4): 483-507.
- Fearon, James D. 2011. "Governance and Civil War Onset." Washington, DC.: World Bank. Accessed 16 February 2014. <https://openknowledge.worldbank.org/handle/10986/9123> License: CC BY 3.0 Unported.
- Fearon, James D., and David D. Laitin. 2003. "Ethnicity, Insurgency, and Civil War." *American Political Science Review* 97 (1): 75-90.
- Fjelde, Hanne. 2009. "Buying Peace? Oil Wealth, Corruption and Civil War 1985-99." *Journal of Peace Research* 47 (2): 199-218.
- Fjelde, Hanne, and Indra de Soysa. 2009. "Coercion, Co-optation or Cooperation? State Capacity and the Risk of Civil War, 1961-2004." *Conflict Management and Peace Science* 26 (1): 5-25.
- Fjelde, Hanne, and Håvard Hegre. 2007. "Democracy Depraved. Corruption and Institutional Change 1985-2000." Paper presented to the 48th Annual Convention of the International Studies Association, Chicago, February 28–March 3.
- Freedom House. 2010. *Freedom in the World 2010. The Annual Survey of Political Rights and Civil Liberties*. Lanham, MD: Rowman & Littlefield and Freedom House.

- Gates, Scott, Håvard Hegre, Mark P. Jones, and Håvard Strand. 2006. "Institutional Inconsistency and Political Instability: Polity Duration, 1800–2000." *American Journal of Political Science* 50 (4): 893–908.
- Geddes, Barbara. 1999. "What Do We Know About Democratization after Twenty Years?" *Annual Review of Political Science* 2:115–44.
- Gleditsch, Kristian S. 2002. "Expanded Trade and GDP Data." *Journal of Conflict Resolution* 46 (5): 712–24.
- Gleditsch, Nils Petter, Peter Wallensteen, Mikael Eriksson, Margareta Sollenberg, and Håvard Strand. 2002. "Armed Conflict 1946–2001: A New Dataset." *Journal of Peace Research* 39 (5): 615–37.
- Gurr, Ted. 1968. "A Causal Model of Civil Strife: A Comparative Analysis Using New Indices." *American Political Science Review* 62 (4): 1104–24.
- Gwartney, James, Joshua Hall, and Robert Lawson. 2010. *Economic Freedom of the World. 2010 Annual Report*. Vancouver, British Columbia: Fraser Institute.
- Harbom, Lotta, and Peter Wallensteen. 2010. "Armed Conflicts, 1946–2009." *Journal of Peace Research* 47 (4): 501–9.
- Hegre, Håvard. 2003. *Disentangling Democracy and Development as Determinants of Armed Conflict*. Working Paper 24637. Washington, DC: World Bank.
- Hegre, Håvard. 2014. "Democracy and Armed Conflict." *Journal of Peace Research* 51 (2): 1–14.
- Hegre, Håvard, Tanja Ellingsen, Scott Gates, and Nils Petter Gleditsch. 2001. "Toward a Democratic Civil Peace? Democracy, Political Change, and Civil War, 1816–1992." *American Political Science Review* 95 (1): 33–48.
- Hegre, Håvard, and Hanne Fjelde. 2009. "Post-conflict Democracy and Conflict Recurrence." In *Peace and Conflict*, edited by J. Joseph Hewitt, Jonathan Wilkenfeld, and Ted Robert Gurr, 79–90. Boulder, CO: Paradigm.
- Hegre, Håvard, Lisa Hultman, and Håvard Møkleiv Nygård. 2011. "Simulating the Effect of Peacekeeping Operations, 2010–2035." In *Social Computing, Behavioral-cultural Modeling and Prediction*, edited by John Salerno, Shanchieh Jay Yang, Dana Nau, and Sun-Ki Chai, 325–34. Lecture Notes in Computer Science 6589. New York: Springer.
- Hegre, Håvard, and Nicholas Sambanis. 2006. "Sensitivity Analysis of Empirical Results on Civil War Onset." *Journal of Conflict Resolution* 50 (4): 508–35.
- Helliwell, John F. 1994. "Empirical Linkages between Democracy and Economic Growth." *British Journal of Political Science* 24 (2): 225–48.
- Hendrix, Cullen S. 2010. "Measuring State Capacity: Theoretical and Empirical Implication for the Study of Civil Conflict." *Journal of Peace Research* 47:271–85.
- Honaker, James, and Gary King. 2010. "What To Do about Missing Values in Time-series Cross-section Data." *American Journal of Political Science* 54 (3): 561–81.
- Kanbur, Ravi, and Xiaobo Zhang. 2005. "Fifty Years of Regional Inequality in China: A Journey through Central Planning, Reform, and Openness." *Review of Development Economics* 9 (1): 87–106.
- Kaufmann, Daniel, Aart Kray, and Massimo Mastruzzi. 2010. *The Worldwide Governance Indicators. Methodology and Analytical Issues*. World Bank Policy Research Working Paper 5430, The World Bank, Washington, DC.

- King, Gary, Michael Tomz, and Jason Wittenberg. 2000. "Making the Most of Statistical Analyses: Improving Interpretation and Presentation." *American Journal of Political Science* 44 (2): 347-61.
- Knutsen, Carl Henrik. 2011. "The Economic Effects of Democracy and Dictatorship." PhD thesis, Department of Political Science, University of Oslo, Oslo, Norway.
- Le Billon, Philippe. 2003. "Buying Peace or Fuelling War: The Role of Corruption in Armed Conflicts." *Journal of International Development* 15:413-26.
- Lipset, Seymour Martin. 1959. "Some Social Requisites of Democracy: Economic Development and Political Legitimacy." *American Political Science Review* 53 (1): 69-105.
- Little, Roderick, J. A., and Donald B. Rubin. 2002. *Statistical Analysis with Missing Data*. 2nd ed. Hoboken, NJ: John Wiley.
- Maddison, Angus. 2007. *Contours of the World Economy 1–2030. Essays in Macroeconomic History*. Oxford, England: Oxford University Press.
- Marshall, Monty G. n.d. "Polity IV Project: Political Regime Characteristics and Transitions, 1800–2009." Accessed January 16, 2014. <http://www.systemicpeace.org/polity/polity4.htm>.
- Mauro, Paolo. 1995. "Corruption and Growth." *The Quarterly Journal of Economics* 110 (3): 681-712.
- Moore, Will H. 1998. "Repression and Dissent: Substitution, Context, and Timing." *American Journal of Political Science* 42 (3): 851-73.
- Muller, Edward N., and Erich Weede. 1990. "Cross-national Variations in Political Violence: A Rational Action Approach." *Journal of Conflict Resolution* 34 (4): 624-51.
- Newey, Whitney. 1987. "Efficient Estimation of Limited Dependent Variables Models with Endogenous Explanatory Variables." *Journal of Econometrics* 36:231-50.
- Organski, A. F. K., and Jacek Kugler. 1980. *The War Ledger*. Chicago: University of Chicago Press.
- Østby, Gudrun. 2008. "Polarization, Horizontal Inequalities and Violent Civil Conflict." *Journal of Peace Research* 45 (2): 143-62.
- PRS Group. 2010. "International Country Risk Guide Methodology." *Political Risk Group*. Accessed January 16, 2014. <http://www.prsgroup.com/PDFS/icrgmethodology.pdf>.
- Przeworski, Adam, Michael E. Alvarez, José Antonio Cheibub, and Fernando Limongi. 2000. *Democracy and Development. Political Institutions and Well-being in the World, 1950–1990*. Cambridge, England: Cambridge University Press.
- Råby, Nils, and Jan Teorell. 2010. "A Quality of Government Peace? Bringing the State Back into the Study of Inter-state Armed Conflict." QOG Working Paper Series 2010:20, The Quality of Government Institute, Göteborg, Sweden. Accessed January 16, 2014. <https://www.qog.pol.gu.se/digitalAssets/1350/1350164201020rabyteorell.pdf>.
- Taydas, Zeynep, Dursun Peksen, and Patrick James. 2010. "Why Do Civil Wars Occur? Understanding the Importance of Institutional Quality." *Civil Wars* 12 (3): 195-217.
- United Nations. 2007. *World Population Prospects. The 2006 Revision*. New York: United Nations.
- Vanhanen, Tatu. 2000. "A New Dataset for Measuring Democracy, 1810-1998." *Journal of Peace Research* 37 (2): 251-65.

- Walter, Barbara. 2004. "Does Conflict Beget Conflict? Explaining Recurring Civil War." *Journal of Peace Research* 41 (3): 371-88.
- Wintrobe, Ronald. 1998. *The Political Economy of Dictatorship*. Cambridge: Cambridge University Press.
- Wood, Reed M., and Mark Gibney. 2010. "The Political Terror Scale (PTS): A Re-introduction and a Comparison to CIRI." *Human Rights Quarterly* 32:367-400.
- World Bank. 2009. "Country Policy and Institutional Assessments—2009 Assessment Questionnaire." Accessed January 16, 2014. <http://siteresources.worldbank.org/IDA/Resources/73153-1181752621336/CPIA09CriteriaB.pdf>.
- World Bank. 2011. *World Development Indicators*. Washington, DC: World Bank.
- Zakaria, Fareed. 1997. "The Rise of Illiberal Democracy." *Foreign Affairs* 76:22-43.